温州肯恩大学
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GMBA 5885 W04
Strategic Mgt Global Info Sys

Introduction to MIS

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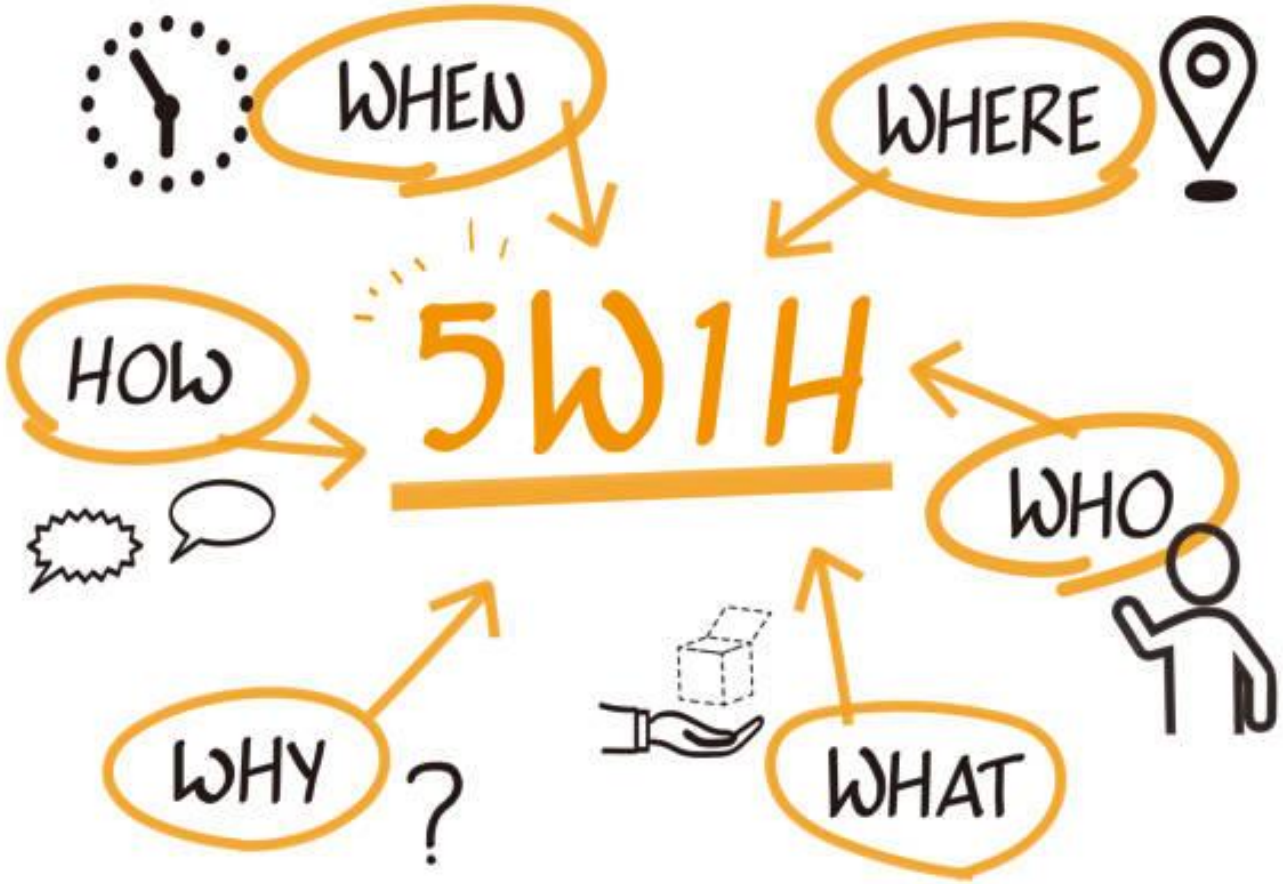
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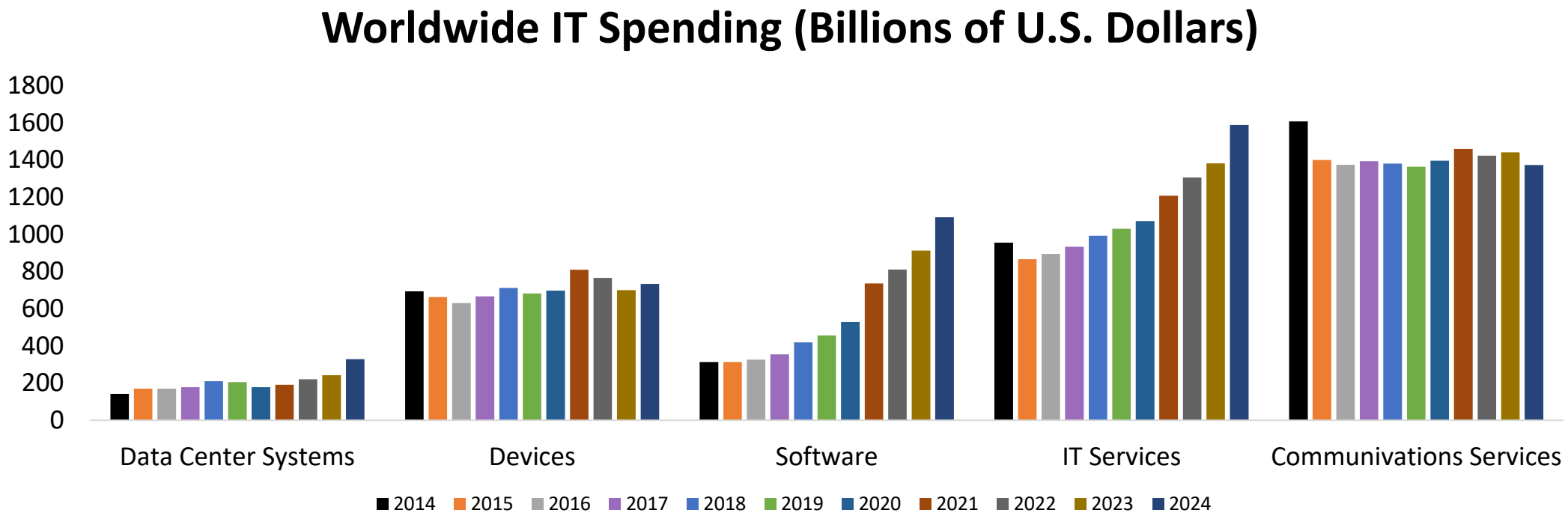
Learning Objectives

By the end of this session, you should be able to:

- Explain how information systems (IS) transform business
- Understand why should managers need to learn IS
- Identify key business objectives supported by IS
- Define an IS and explain how it works
- Describe organizational, management, and technology dimensions of IS
- Recognize key challenges and opportunities from emerging IS technologies
- Describe major approaches to studying IS

IS Transforms Business



Source: Gartner

"If a business isn't using information systems connected to the Internet and mobile apps, chances are it isn't being as effective as it could be."

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IS Transforms Business

- In today's digital economy, **data** is as critical as capital or labor.
- IS provides structure and intelligence to **how data becomes value**.
- Technology-driven changes are transforming jobs and careers
  - How you work, where you work, and how well you will be compensated
- Many millions of new jobs have been created
  - Not only for IT specialists but also for business professionals who use IT tools
- Gaining a deeper understanding of how technology works and is evolving is essential (e.g., AI tools).

"Every global company is becoming a tech company … we're seeing technology as a critical component for business success."
– Jeremy King, CTO@Walmart Labs

Why Managers Should Participate in IS Decisions

Reason
IS must be managed as a critical resource since it permeates almost every aspect of business.
IS enable change in the way people work both inside and outside of the enterprise.
IS are at the heart of integrated Internet-based or mobile solutions that are replacing standard business processes.
IS enable or inhibit business opportunities and new strategies.
IS can be used to combat business challenges from competitors.
IS enable customers to have greater pull on businesses and communities by giving them new options for voicing their concerns and opinions using social media.
IS can support data-driven decision making.
IS can help ensure the security of key assets.

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Skills for Successful IT Use by Managerial Role

Managerial Role	Skills
Visionary	Creativity, curiosity, confidence, focus on business solutions, flexibility
Informational and Interpersonal	Communication, listening, information gathering, interpersonal skills
Structural	Project management, analytical, organizational, planning, leading, controlling

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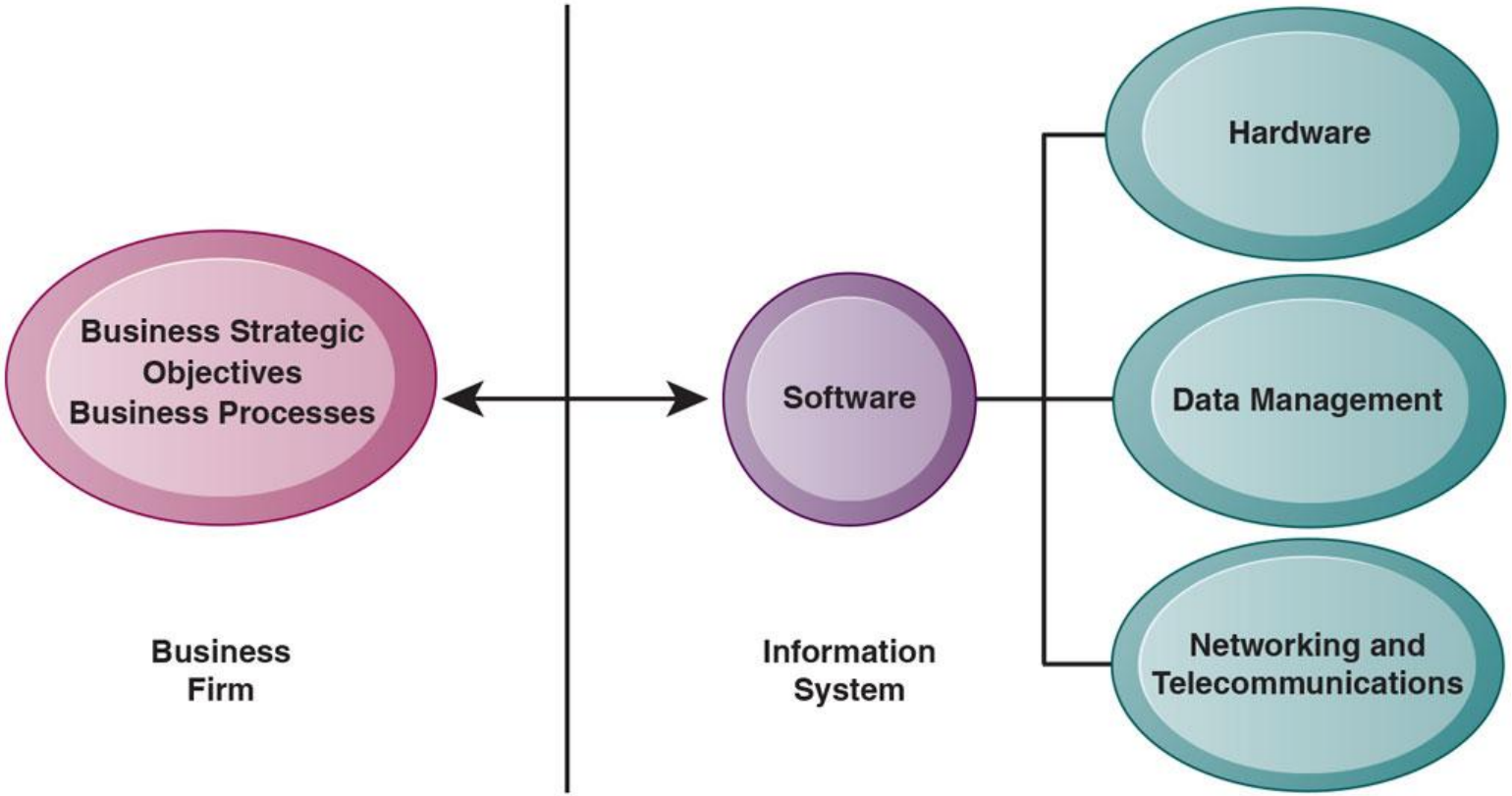
Business Objectives Supported by IS

Interdependence of Organization and Information Systems



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Business Objectives Supported by IS

Firms invest heavily in information systems to achieve **6** strategic business objectives:

- Operational excellence
- New products, services, and business models
- Customer and supplier intimacy
- Improved decision making
- Competitive advantage
- Survival

Quiz

Which of the following is **not** an important business objective driving investment in information systems and technologies?

A. Operational excellence

B. Improved decision making

C. Improved employee benefits

D. New products, services, and business models

What Is an IS

- Information Technology (IT)
  - The **hardware** and **software** a business uses to achieve objectives
- Management Information System (MIS/IS)
  - A set of interrelated components that collect (or retrieve), process, store, and **distribute information** to
    - ❑Support decision making, coordination and control
    - ❑Help analyze problems, visualize complex issues, and develop new products or services

What Is an IS

Comparison of Data, Information, and Knowledge

	Data	Information	Knowledge
Definition	Simple observations of the state of the world	Data endowed with relevance and purpose	Information from the human mind (includes reflection, synthesis, context)
Characteristics	• Easily structured • Easily captured on machines • Often quantified • Easily transferred • Mere facts	• Requires unit of analysis • Data that have been processed • Human mediation necessary	• Hard to structure • Difficult to capture on machines • Often tacit • Hard to transfer
Example	Daily inventory report of all items	Daily inventory report of items with low levels of stock	Report of which items need to be reordered after considering inventory levels, anticipated labor strikes, and a flood affecting a supplier

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What Is an IS

Information Characteristics Across Levels

	Top Management	Middle Management	Supervisory and Lower-Level Management
Time Horizon	Long: years	Medium: weeks, months, years	Short: day to day
Level of Detail	Highly aggregated Less accurate More predictive	Summarized Integrated Often financial	Very detailed Very accurate Often nonfinancial
Source	Primarily external	Primarily internal with limited external	Internal
Decision	Extremely judgmental Uses creativity and analytical skills	Relatively judgmental	Heavy reliance on rules

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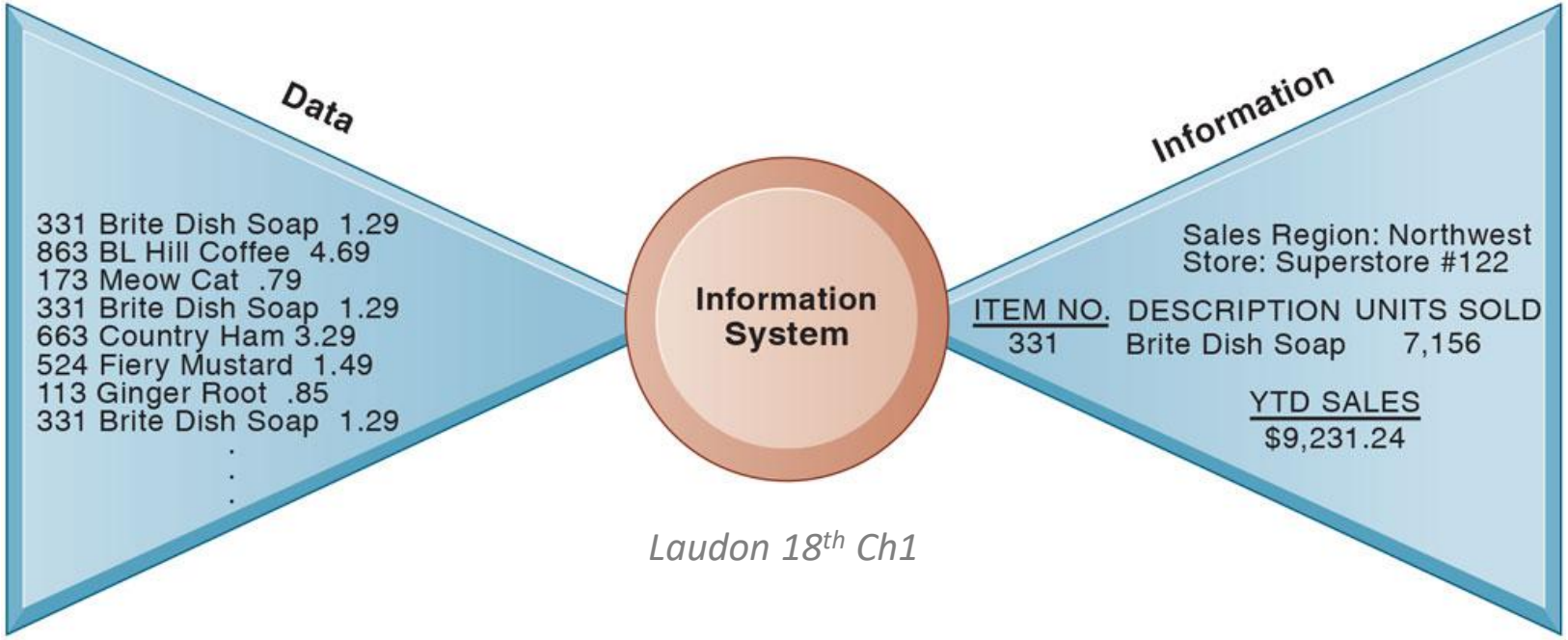
What Is an IS

Example: Raw data from a *supermarket checkout counter* can be processed and organized to produce meaningful information.



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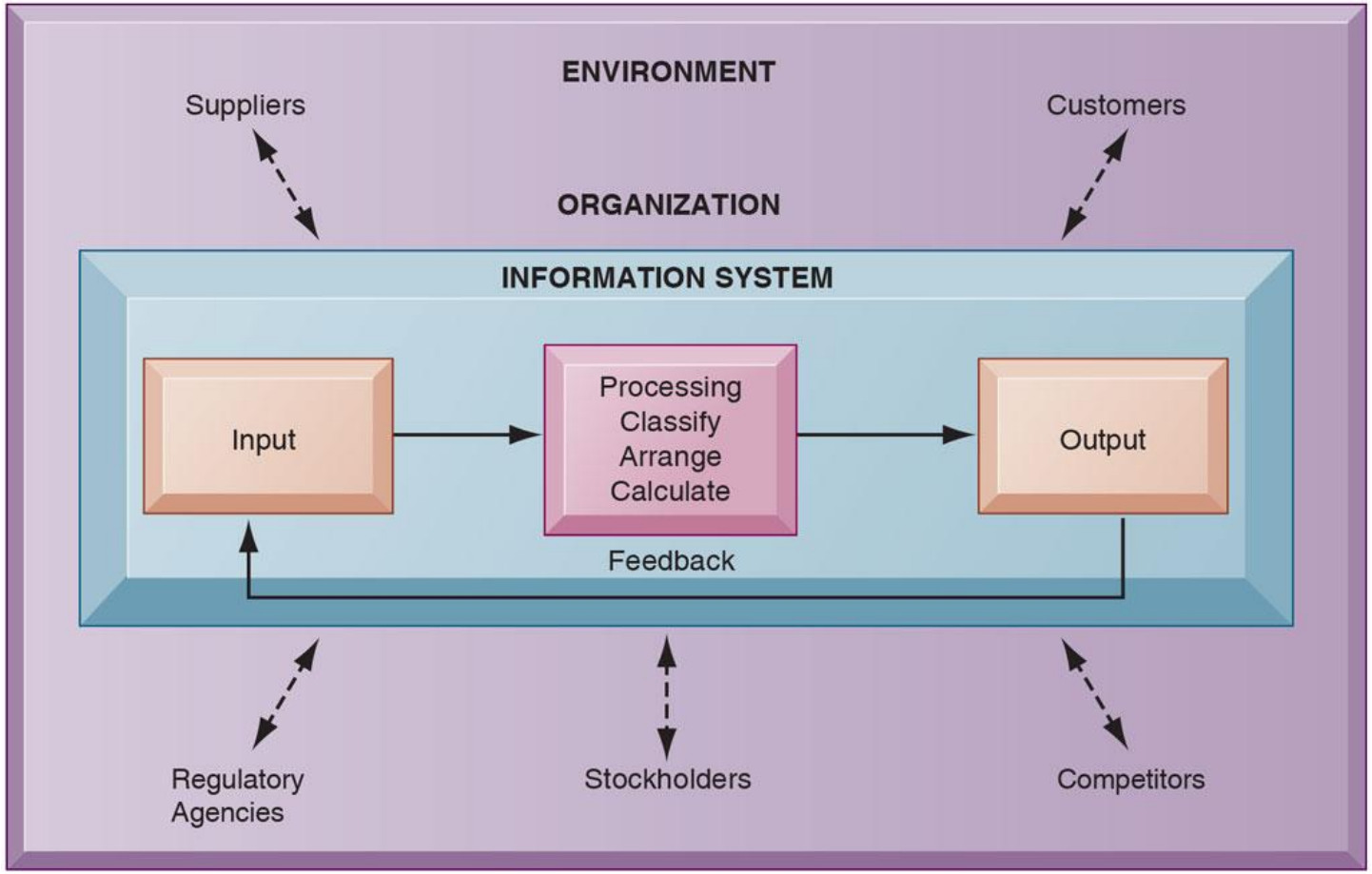
Streams of raw facts

Data shaped into meaningful, useful form

How an IS Works

Functions of an IS



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How an IS Works

- Three Core Activities
  - **Input:** Captures raw data from inside or outside the organization
  - **Processing:** Converts data into meaningful information
  - **Output:** Distributes processed information to users and decision-makers
- Feedback
  - Output returned to relevant people or systems to improve future inputs and processes
- External Actors
  - Customers, suppliers, competitors, stockholders, and regulators continuously interact with the system

Quiz

Processing involves which of the following?

A. Returning output to appropriate members of the organization to help them evaluate or correct the input stage.

B. Transferring information to the people who will use it or to the activities for which it will be used.

C. Capturing or collecting raw data from within the organization or from its external environment.

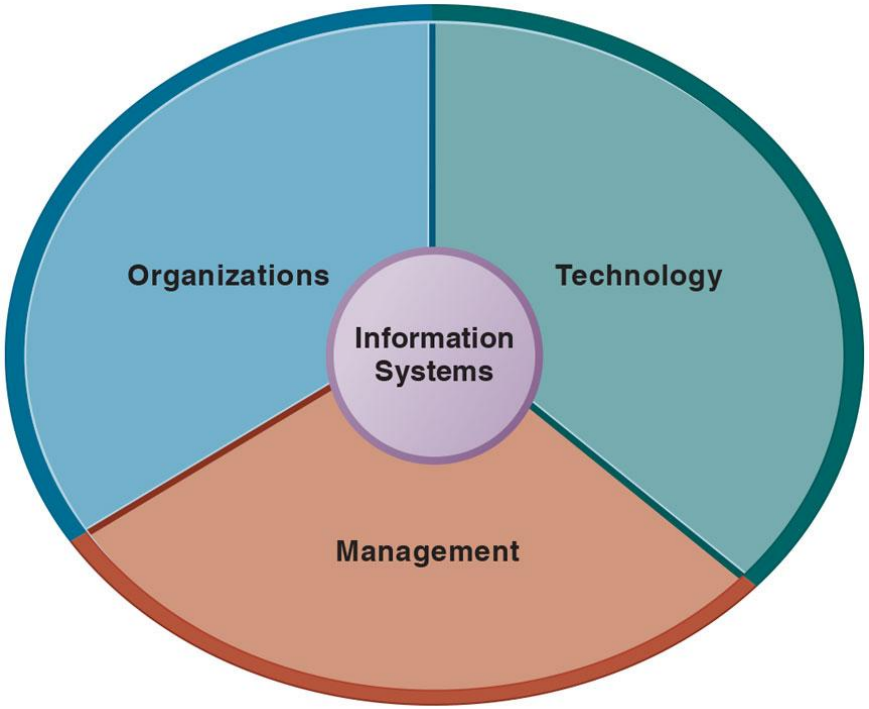
D. Converting data into a meaningful form by classifying, arranging, and performing calculations on it.

IS Dimension

To fully understand IS:

- Consider the **broader** organizational, managerial, and information technology dimensions of IS
- Recognize how IS provides **solutions** to business challenges and problems

Information systems are more than computers!



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IS Dimension 1: Organization

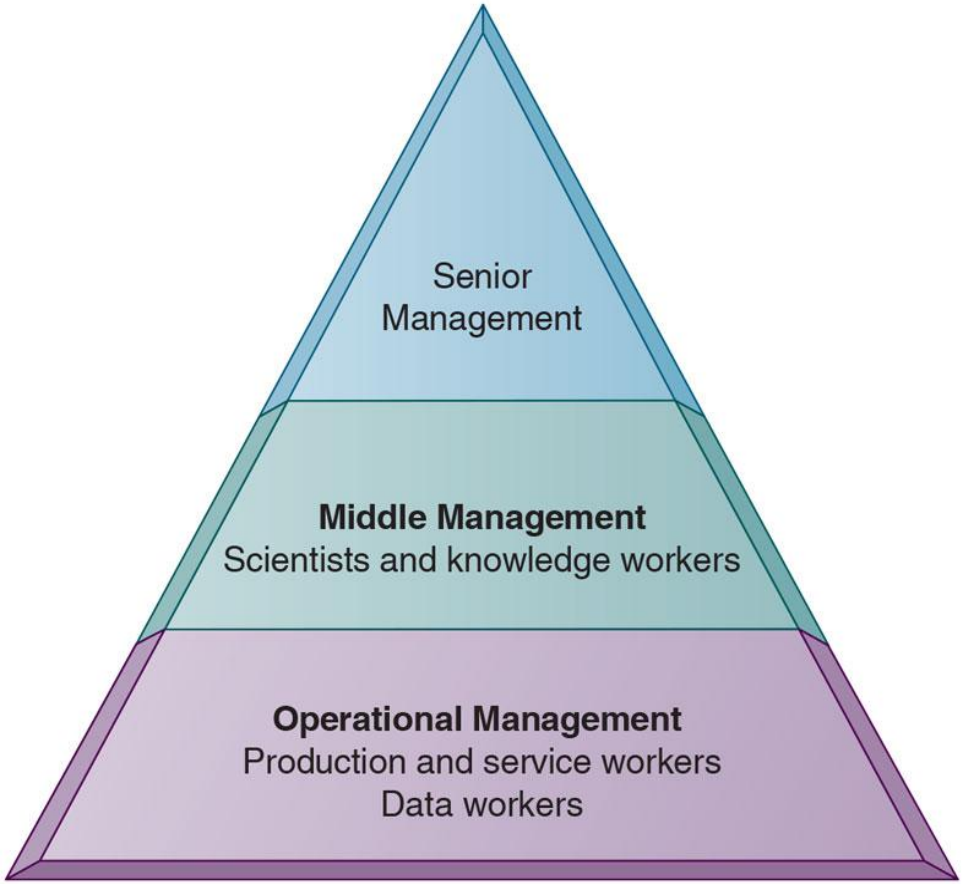
IS and organizations shape each other.

- IS support key functions (marketing, finance, HR) and processes (ordering, hiring, product development).
- Organizational rules and practices are often built into and automated by IS.

❑Culture affects how IS are adopted and used.

❑Different roles/interests create politics that influence IS design and implementation.

Levels in a Firm



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IS Dimension 2: Management

- Managers **interpret challenges**, make decisions, and set strategy for the organization.
- They **allocate resources** and coordinate people, processes, and technology.
- Management requires not only control and planning, but also **creativity** – developing new products, services, and ways of working.
- Effective management depends on **timely, accurate information** – exactly **what IS provide**.
- IS enable managers to analyze problems, generate insights, and design innovative solutions.

IS Dimension 3: Technology

- The technology dimension covers the **tools** and **platforms** that enable IS.
- Key Components
  - **Hardware:** devices for input, processing, storage, and output
  - **Software:** programs that control and coordinate hardware
  - **Data management:** tools for storing, organizing, and accessing data
  - **Networking & telecommunications:** technologies that connect systems and enable data exchange
- IT infrastructure = Technologies + People → foundation on which IS are built

Quiz

Which of the following makes **long-range** strategic decisions in a firm?

A. Senior management

B. Knowledge workers

C. Middle management

D. Operational management

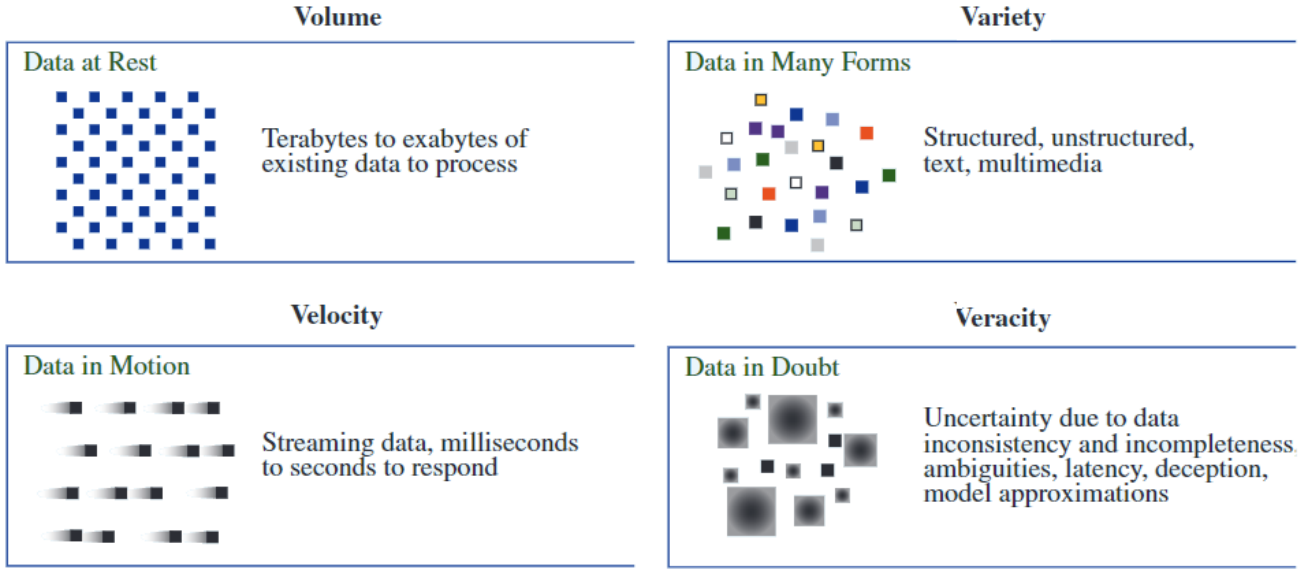
Key Technologies Shaping Modern IS

1) Internet of Things (IoT)

- Network of connected devices with sensors and software.
- Generates continuous data used for monitoring, automation, and prediction.
- Major source of big data.

2) Big Data

- Extremely large, fast, and varied data (4v: volume, velocity, variety, velocity).
- Enables deeper insights, forecasting, and optimization beyond traditional data tools.



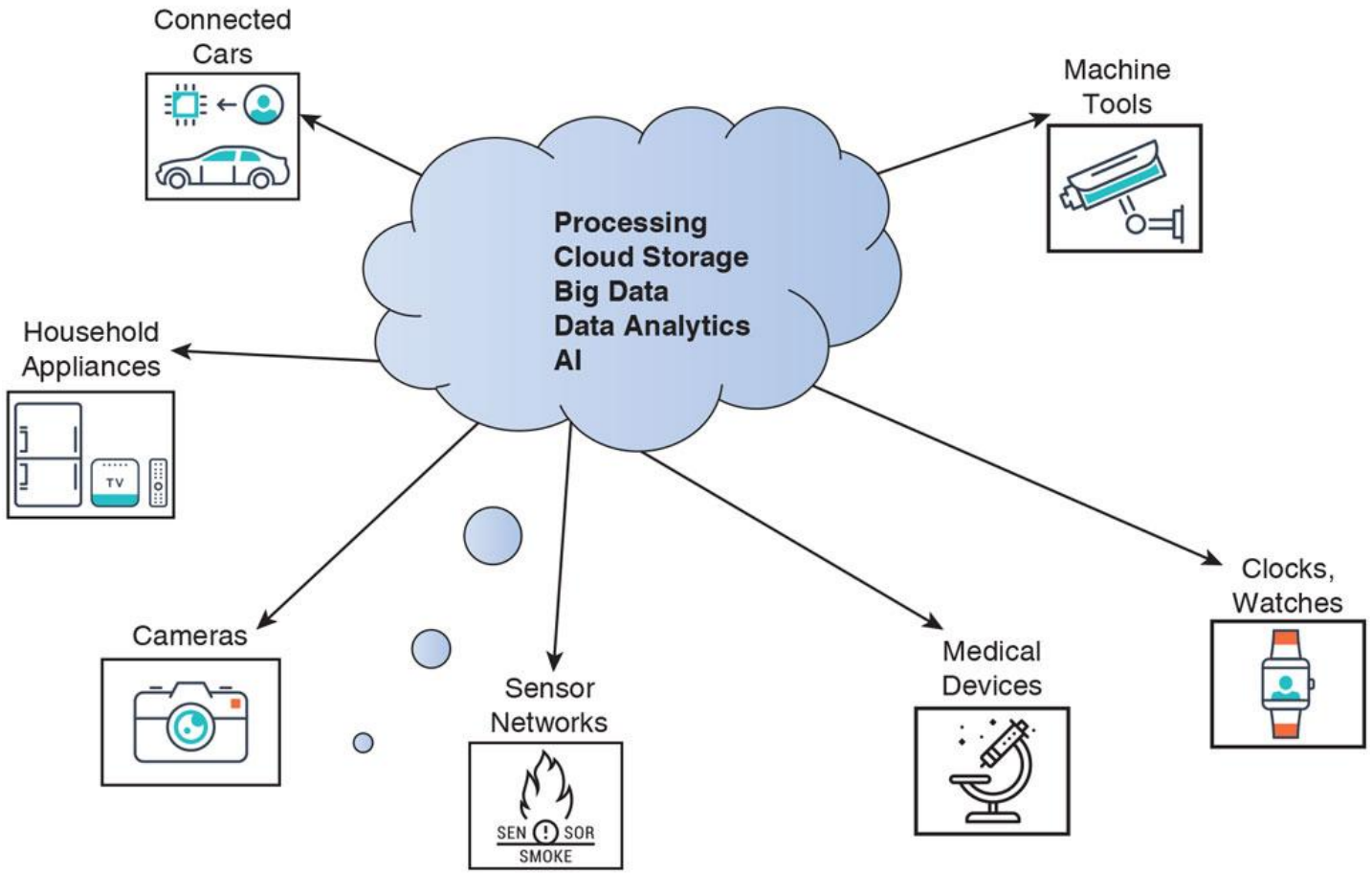
Key Technologies Shaping Modern IS

3) Cloud Computing
   - On-demand access to processing, storage, and software via the Internet
   - Provides scalable infrastructure needed for big data and AI workloads

4) Artificial Intelligence (AI)
   - Systems that perform tasks requiring human intelligence (e.g., recognition, prediction, decision-making)
   - Machine learning, deep learning, generative AI (e.g., ChatGPT, Gemini, Copilot)
   - Driving transformation across industries

Key Technologies Shaping Modern IS

IoT → generates data → **Big Data** → processed in the **Cloud** → powering **AI** insights and decisions



Complementary Assets

IS alone does not create value — returns come when technology is combined with the right organizational and managerial conditions.

1) IS creates value when it improves:
   - Productivity
   - Revenue and market position
   - Decision-making and business processes
2) But technology is not enough
   - Firms with the same IT investment often get very different results.
   - Success depends on complementary assets that support and enable the technology.

Complementary Assets

IS alone does not create value — returns come when technology is combined with the right organizational and managerial conditions.

3) Key complementary assets
   - **Organizational:** effective culture, business model, efficient processes, strong IS team
   - **Managerial:** leadership support, incentives, collaboration, training
   - **Social:** infrastructure, education, standards, regulations, partner ecosystems

4) Takeaway
   - High returns require **aligning** technology with organizational change, management capability, and supporting environment.
   - Firms that invest in both **IT + complementary assets** gain superior performance.

Quiz

Which of the following is **not** an example of a **complementary social asset** required to optimize returns from information technology investments?

A. Government and private sector standards

B. IT-enriched educational programs

C. Laws and regulations creating fair, stable market environments

D. Efficient business processes

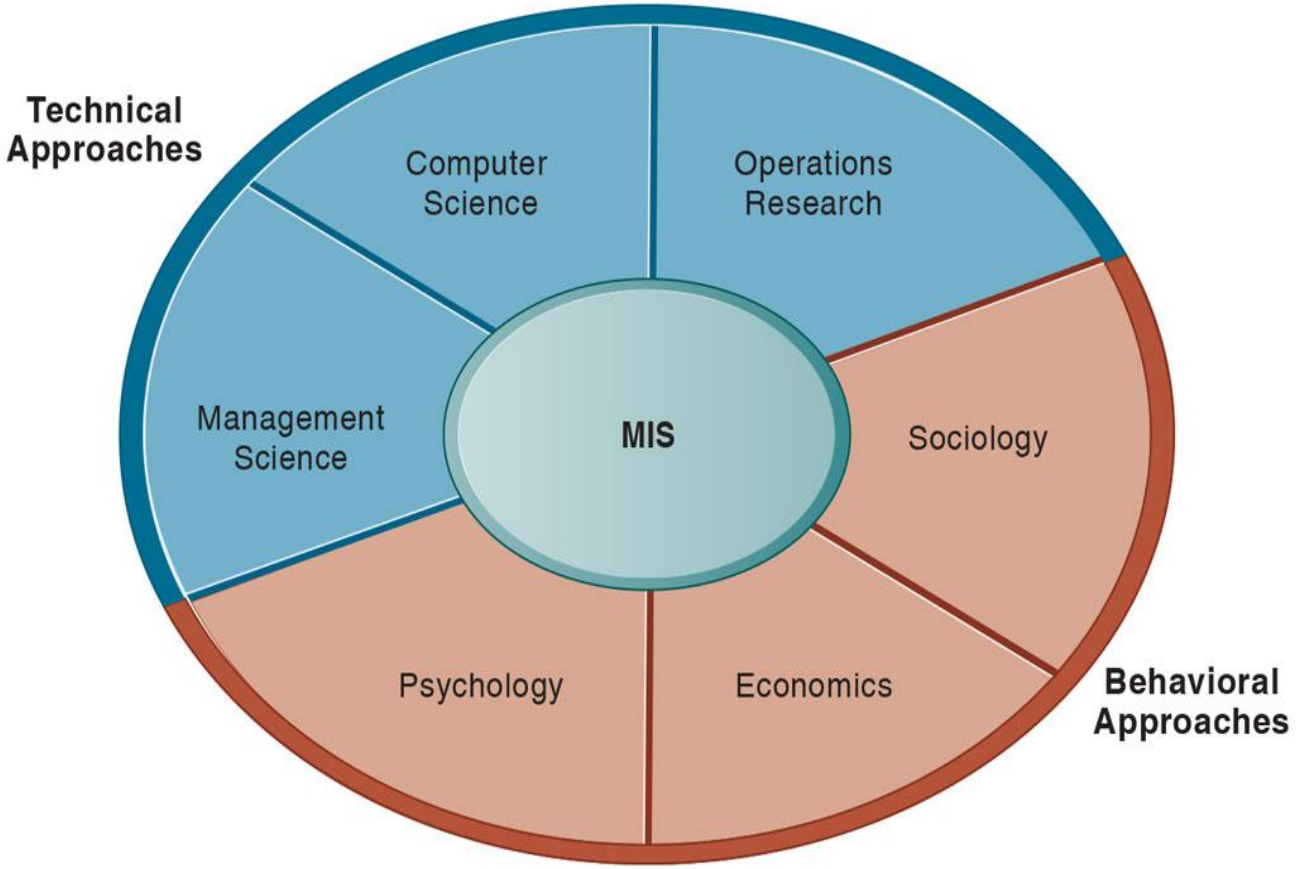
Research Field – MIS

The field of Management Information Systems (MIS):

- Tries to achieve broader information systems literacy
- MIS deals with behavioral issues as well as technical issues surrounding the **development**, **use**, and **impact** of information systems that managers and employees in the firm use

- The study of MIS is a **multidisciplinary field**.
- No single theory or perspective dominates.

Contemporary Approaches to IS



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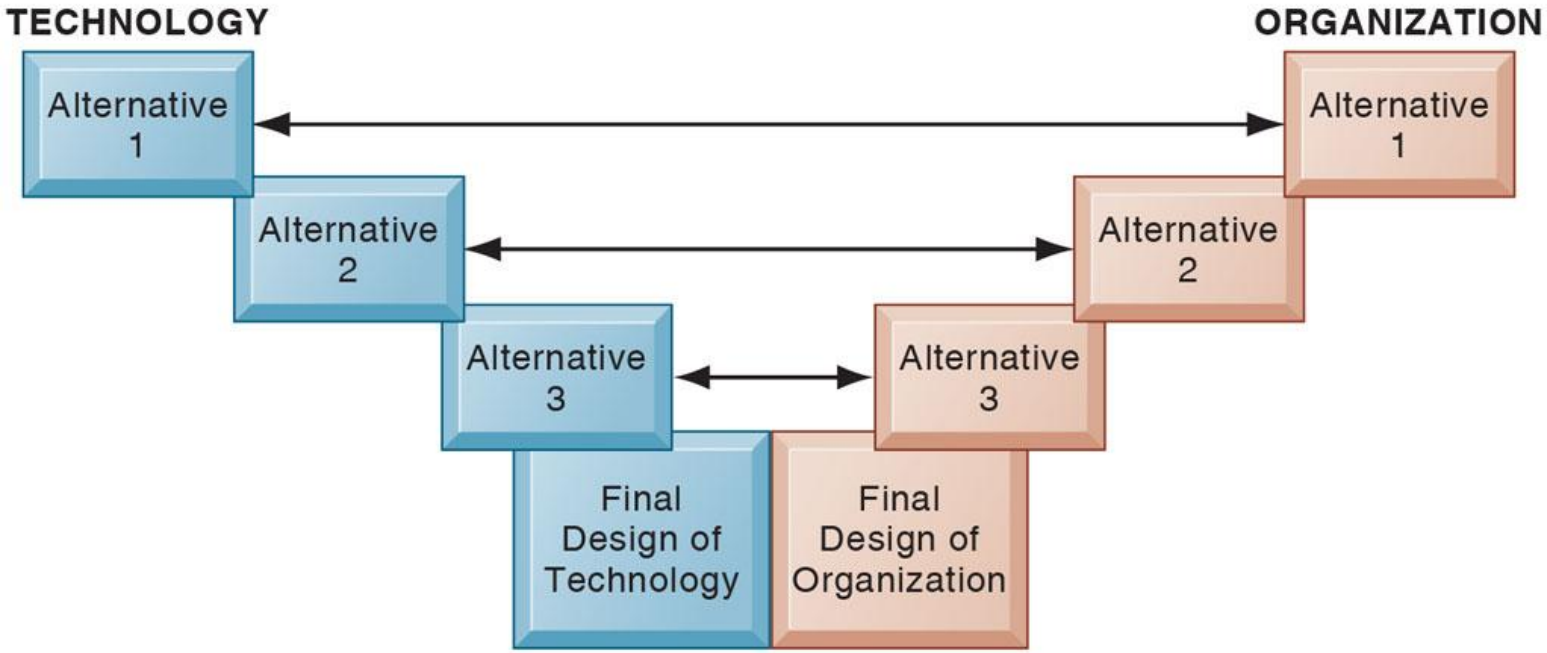
Research Topics in MIS

Research Topic in MIS (MIS Quarterly Research Curations)

- Information Systems Development
- IS Sourcing
- IT Project Management
- Data Management
- Information Systems Alignment
- IS Use
- Knowledge Management
- Health Information Technology
- IT-supported Collaboration
- Online Word-of-Mouth
- Platforms and Ecosystems
- IT Workforce
- Information Privacy
- IS Control & Governance
- Societal Implications of ICTs

A Sociotechnical Perspective on MIS

Optimal organizational performance is achieved by jointly optimizing both the **social** and **technical** systems used in production.



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Key Challenges in New MIS Technologies

Technology Changes	Business Challenges
Cloud Computing Organizations move IT operations to cloud-based platforms for scalability and cost efficiency.	How much IT infrastructure should move to the cloud? What are the trade-offs in cost, control, and security?
Big Data & IoT Firms collect vast data from sensors, web traffic, and social media to improve decisions.	Does your firm have the tools and talent to analyze and leverage big data effectively?
Artificial Intelligence (AI) Systems perform tasks requiring human-like intelligence—recognizing patterns, predicting, learning.	Where can your company apply AI? What benefits, costs, and expertise are needed for implementation?
Mobile Platforms Smartphones, tablets, and wearables enable collaboration and connectivity anywhere.	Is your firm maximizing mobile capabilities for employees and customers? How can it enhance value?

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Key Challenges in New MIS Technologies

Management	Business Challenges
Return on Investment (ROI) Firms invest heavily in information systems but often struggle to measure the value gained.	How can your firm accurately assess the benefits of IT investments and identify more cost-effective alternatives?
Online Collaboration & Social Networks Platforms like Google Workspace, Microsoft 365, Slack, and Zoom reshape teamwork and communication.	Is your firm effectively using digital tools to improve coordination and knowledge sharing? Which platforms add the most value?

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Key Challenges in New MIS Technologies

Organizations	Business Challenges
Security & Privacy Data protection and privacy have become critical public and regulatory concerns.	How secure is your company's data? Are its privacy policies adequate and compliant with evolving laws?
Social Business Firms leverage social platforms (Facebook, X, TikTok, internal networks) to engage employees and customers.	Is your firm gaining measurable value from social business tools? How could engagement be deepened?
Remote & Hybrid Work Cloud and mobile technologies enable flexible work models across locations.	What policies and technologies will shape your firm's future of work? What challenges and opportunities exist?

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Summary

- IS transforms business by enabling data-driven operations, new business models, and competitive advantage.
- IS support 6 core strategic business objectives: operational excellence, new products, services, and business models, customer and supplier intimacy, improved decision making, competitive advantage, survival.
- An IS **collects**, **processes**, **stores**, and **distributes** data to support decision-making.
- Fully understanding IS requires considering the organizational, managerial, and technological dimensions.
- Key technologies – IoT, Big Data, Cloud Computing, and AI – are reshaping how firms operate and compete.
- High value from IT comes when firms invest in complementary assets (culture, leadership, processes, skills).
- MIS is a multidisciplinary field integrating technical and behavioral perspectives.